

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4303

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COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- Identify key components required in a navigation bar for an e-commerce website.
- Design a responsive navigation bar using appropriate CSS techniques.
- Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
- Evaluate the usability and suggest improvements for better user experience.

Ans:

(a) Key Components

- Website logo
- Navigation links (Home, Products, Categories, Cart, Contact)
- Search bar
- User account/login
- Shopping cart icon

(b) Design

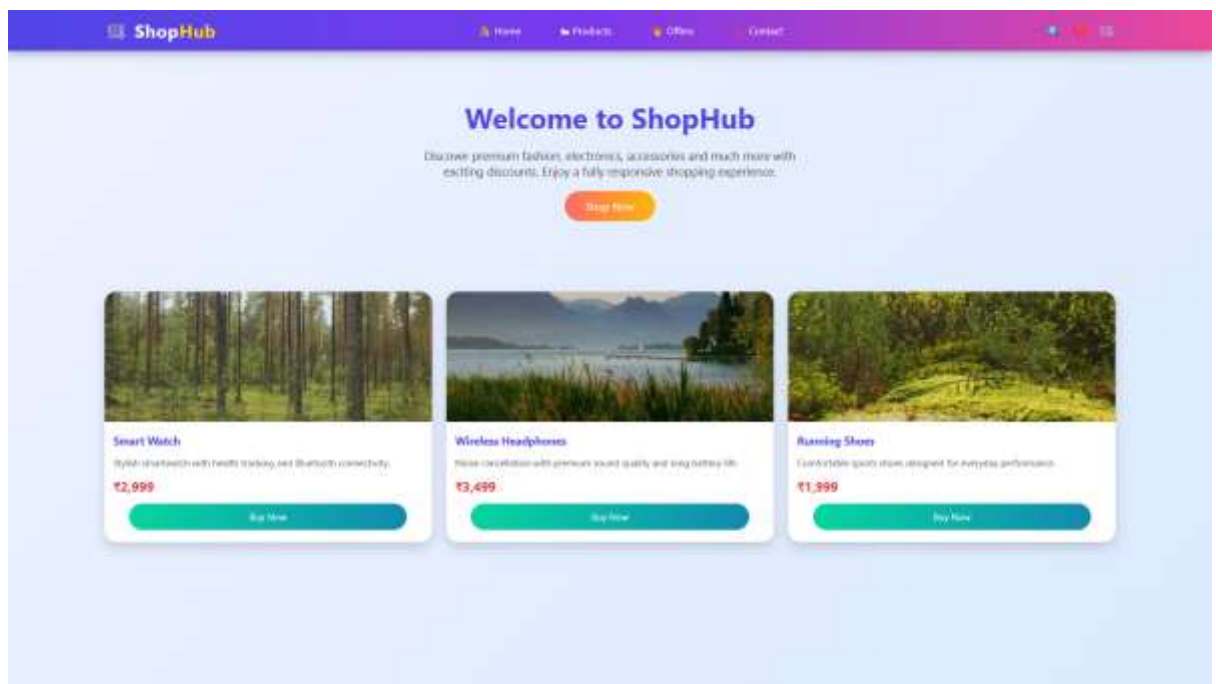
- Use HTML for the navigation structure.
- Use CSS Flexbox for alignment.
- Apply gradient background, hover effects, and spacing.

(c) Responsive Adaptation

- Use media queries for smaller screens.
- Hide navigation links and display a hamburger menu.
- Toggle the menu using JavaScript.

(d) Evaluation

- Responsive on all devices.
- Easy navigation and improved accessibility.
- Smooth animations enhance user experience.



Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- Identify suitable CSS properties for hover effects.
- Suggest appropriate CSS animation or transition for product interaction.
- Design a hover effect for a product card.
- Evaluate performance and user experience considerations.

(a) CSS Properties

- transform
- transition
- box-shadow
- opacity
- background-color

(b) CSS Animation/Transition

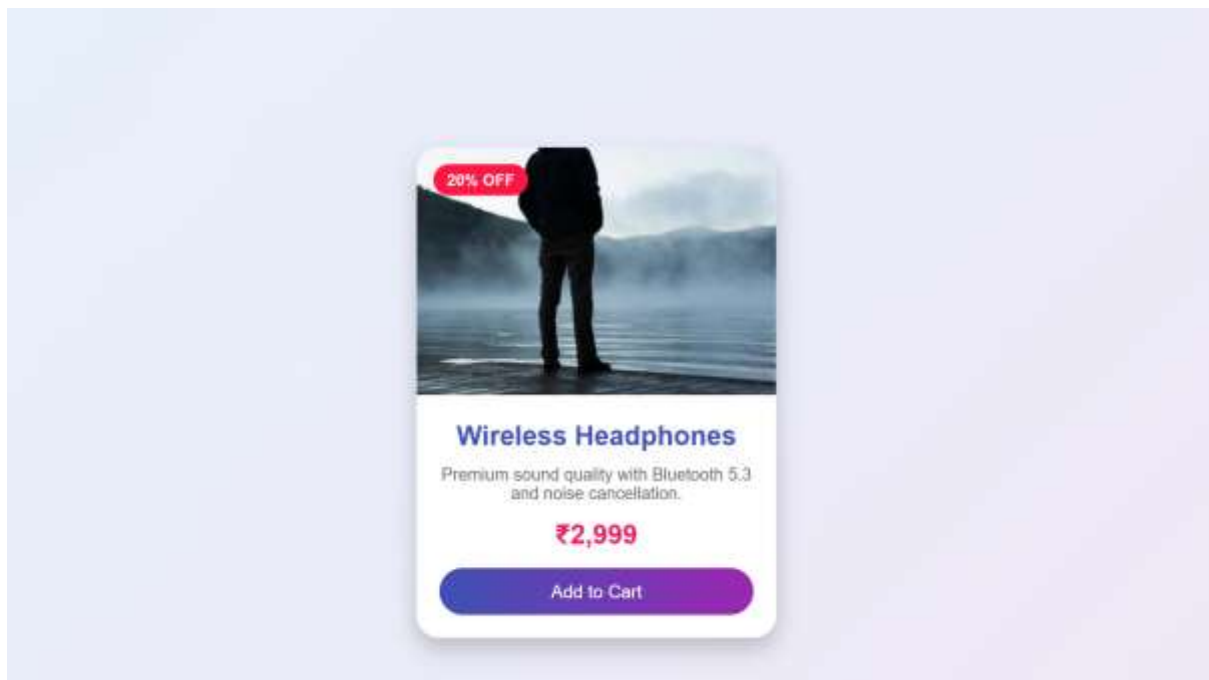
- Use transition: 0.3s ease;
- Apply transform: scale() on hover.
- Add shadow and color transition.

(c) Hover Effect Design

- Zoom the product image.
- Lift the product card.
- Add shadow and highlight the button.

(d) Evaluation

- Makes products more attractive.
- Improves customer interaction.
- Keep animations lightweight for better performance.



Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- Identify layout requirements of a dashboard (rows, columns, alignment).
- Compare Flexbox and Grid based on layout needs.
- Choose the most suitable layout technique and justify your choice.

(d) Evaluate the scalability and responsiveness of the chosen approach.

Ans:(a) Layout Requirements

- Header
- Sidebar
- Main content
- Dashboard cards
- Footer

(b) Comparison

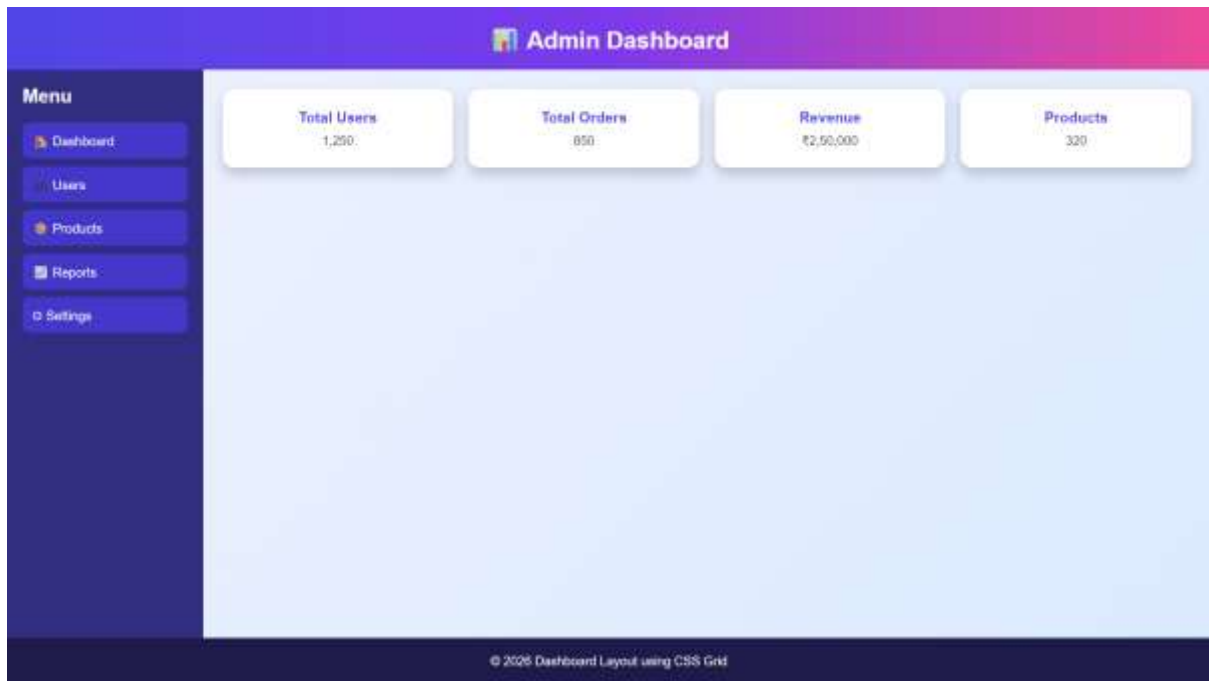
Flexbox	Grid
One-dimensional layout	Two-dimensional layout
Best for rows or columns	Best for complete page layouts
Easy alignment	Better for dashboards

(c) Choice

- CSS Grid is preferred because it efficiently manages rows and columns in dashboard layouts.

(d) Evaluation

- Easy to maintain.
- Highly responsive.
- Suitable for large and complex layouts.



Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- Identify key sections of a blog layout.
- Design a mobile-first layout using CSS.
- Demonstrate how the layout scales for larger screens using media queries.
- Evaluate accessibility and readability of the design.

Ans:(a) Key Sections

- Header
- Navigation
- Blog posts
- Sidebar
- Footer

(b) Design

- Start with a mobile layout.
- Use Flexbox.
- Apply clean typography and spacing.

(c) Scaling

- Use media queries.
- Display sidebar beside content on larger screens.
- Convert navigation to horizontal layout.

(d) Evaluation

- Improves readability.
- Better accessibility.
- Works well across all screen sizes.



Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- Identify appropriate JavaScript events for menu interaction.
- Design event handling logic for menu toggle functionality.
- Implement interaction for user actions (click/hover).
- Evaluate performance and suggest improvements.

Ans:(a) JavaScript Events

- click
- mouseover
- mouseout

(b) Event Handling Logic

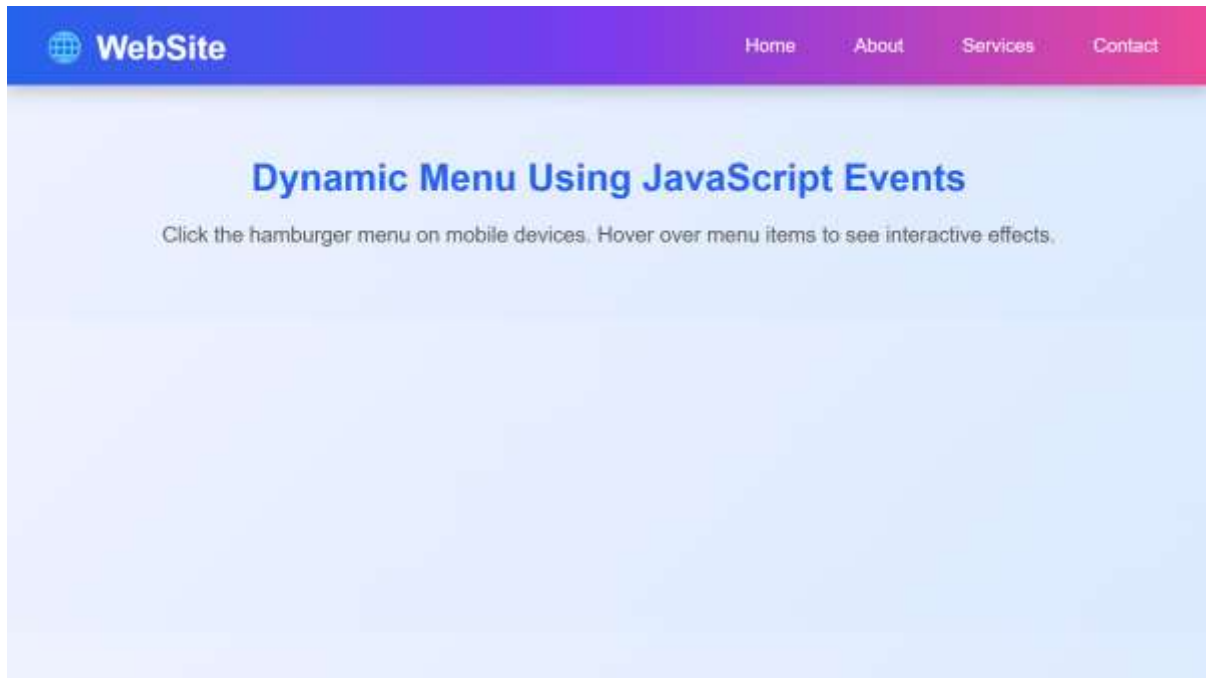
- Detect menu button click.
- Show or hide the navigation menu.
- Update menu state dynamically.

(c) User Interaction

- Click to open/close the menu.
- Hover to highlight menu items.
- Provide smooth visual feedback.

(d) Evaluation

- Improves navigation experience.
- Responsive on mobile devices.
- Use event delegation and optimized event listeners for better performance.



Code of Conduct:

I T. Nikhil Kumar Reddy (192372024) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Nikhil Kumar Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts

		requirements and concepts			
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation